94. Option (d) is correct.

$$\int_0^2 h(x)dx = \int_0^1 |x-1|[x]dx + \int_1^2 |x-1|[x]dx$$

$$= \int_0^1 (1-x).(0)dx + \int_1^2 (x-1).(1)dx$$

$$= \int_1^2 (x-1)dx$$

$$= \left[\frac{x^2}{2} - x\right]_1^2$$

$$= \left(\frac{4}{2}-2\right) - \left(\frac{1}{2}-1\right)$$

$$= \frac{1}{2}$$

Solution for Q.no. 95 and 96:

Given, $\int \frac{dx}{\sqrt{x+1}-\sqrt{x-1}} = \alpha(x+1)^{\frac{3}{2}} + \beta(x-1)^{\frac{3}{2}} + C$ ...(i)

Let $I = \int \frac{dx}{\sqrt{x+1}-\sqrt{x-1}}$

$$= \int \frac{(\sqrt{x+1}+\sqrt{x-1})dx}{(\sqrt{x+1}-\sqrt{x-1})(\sqrt{x+1}+\sqrt{x-1})}$$

$$\Rightarrow \quad I = \frac{1}{2}\int \sqrt{(x+1)}\,dx + \frac{1}{2}\int \sqrt{x-1}\,dx$$

$$\Rightarrow \quad I = \frac{1}{2}.\frac{(x+1)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + \frac{1}{2}.\frac{(x-1)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + C$$

$$\Rightarrow \quad I = \frac{1}{3}(x+1)^{\frac{3}{2}} + \frac{1}{3}(x-1)^{\frac{3}{2}} + C$$

$$\Rightarrow \int \frac{dx}{\sqrt{x+1}-\sqrt{x-1}} = \frac{1}{3}(x+1)^{\frac{3}{2}} + \frac{1}{3}(x-1)^{\frac{3}{2}} + C \quad ...(ii)$$

95. Option (a) is correct.

From (i) and (ii) we get

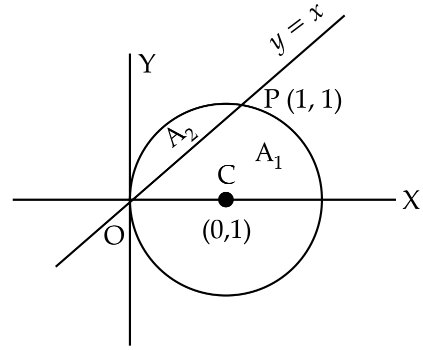
$$\alpha = \frac{1}{3}$$

96. Option (c) is correct.

From (i) and (ii) we get

$$\beta = \frac{1}{3}$$

Solution for Q.no. 97 and 98:



Given, $x^2 + y^2 - 2x = 0$

$\Rightarrow \quad (x-1)^2 + y^2 = 1$

or

$y^2 = 1 - (x-1)^2$

Area of minor segment $(A_2) = \int_0^1 (\sqrt{1-(x-1)^2} - x)dx$

$$\Rightarrow A_2 = \left[\frac{(x-1)}{2}\sqrt{1-(x-1)^2} + \frac{1}{2}\sin^{-1}(x-1) - \frac{x^2}{2}\right]_0^1$$

$$\Rightarrow A_2 = \left[\left(0+\frac{1}{2}(0)-\frac{1}{2}\right) - \left(0+\frac{1}{2}\sin^{-1}(-1)-0\right)\right]$$

$$\Rightarrow A_2 = \frac{\pi-2}{4}$$

97. Option (d) is correct.

Since, area of given circle $= \pi(1)^2 = \pi$

$$\Rightarrow \quad A_1 + A_2 = \pi$$

$$\Rightarrow \quad A_1 = \pi - \frac{\pi-2}{4}$$

$$= \frac{3\pi+2}{4}$$

98. Option (a) is correct.

$$\frac{2(A_1+A_2)}{A_1-3A_2} = \frac{2(\pi)}{\left(\frac{3\pi+2}{4}\right) - \frac{3(\pi-2)}{4}}$$

$$= \pi$$

Solution for Q.no. 99 and 100:

99. Option (d) is correct.

Given, $3f(x) + f\left(\frac{1}{x}\right) = \frac{1}{x} + 1$ ...(i)

Replacing (x) by $\left(\frac{1}{x}\right)$, we get

$$3f\left(\frac{1}{x}\right) + f(x) = x + 1 \quad ...(ii)$$

Now, on $3 \times$ eq. (i) – eq. (ii), we get

$$8f(x) = \frac{3}{x} + 3 - x - 1$$